

DISCUSSION

SESSION 3

P. H. W. WOLFF, *Session Chairman*

FOR the last two days we have heard all about the plans, the science, and the technology of fast breeder reactors. This morning's session will be devoted to design and we have some eminent speakers to give us their thoughts on how fast reactors have been made so far and what they see as their plans for the construction of fast reactors in the future.

Presentation of Papers

YU. N. KOSHKIN—*Paper 3/1*

I shall be describing the design and equipment of the BN-350 reactor but, before going into detail, I want to mention that the work on this reactor was done by a large number of planning, scientific, and design organizations, and that a considerable amount of experimental work was conducted, both in chemistry and engineering, before construction was started.

TABLE D1. MAIN CHARACTERISTICS OF THE BN-350 REACTOR

1. Thermal power	1000 MW	11. Number of core assemblies	211
2. Electrical power	150 MW	12. Total number of assemblies	695
3. Quantity of fresh water produced per day	110,000 tons	13. Number of fuel elements in core assembly	169
4. Primary circuit sodium flow	14,100 t/hr	14. Can for core element	$\varnothing 6.1 \times 0.4$ mm
5. Temperature at entry	300°C	15. Can for radial blanket element	$\varnothing 14.2 \times 0.5$ mm
6. Temperature at exit	500°C	16. Can for axial blanket element	$\varnothing 12 \times 0.4$ mm
7. Height of core	1060 mm	17. Steam temperature	430°C
8. Diameter of core	1500 mm	18. Steam pressure	50 atm.g
9. Mean power density in the core	500 kW/l.		
10. Number of parallel loops	6		

I will now deal with the general aspects of the construction of the reactor (main characteristics given in Table D1). The reactor vessel is made from stainless steel. Inside is a shield to protect the vessel from radiation. Between the shield and the vessel there is a gap through which sodium is passed from the pressure header for cooling the reactor vessel. In the lower part there is a

pressure chamber where the sodium passes through the pipes. From the pressure chamber the sodium is sent through a distributor, which supplies the assemblies in the core as well as in the breeding zone. In the upper part of the reactor are two rotating plugs, each of which has its own electrical system. The smaller plug is located in the larger one. On the smaller plug is the central column containing the reactor's control and safety mechanisms and the refuelling mechanism is also on the smaller plug. Inside the reactor vessel there is a mechanism called the elevator, at the bottom of which is a socket to take one or more assemblies either from the core or from the breeding zone. In the upper part is the transfer mechanism with a sealed container. It is shielded.

The refuelling technology is as follows: with the help of the small and the large plugs the refuelling mechanism is directed towards the assembly with which we are concerned. It is withdrawn and put into the socket of the elevator. With the assistance of the elevator drive this socket moves up until it meets the gripping device of the assembly transfer mechanism. From here the mechanism takes the assembly from the socket and passes it on. Such is the general operating scheme of the refuelling system.

Now a few words about the plug shielding. Each plug has a bismuth-tin seal. The alloy is in a solid condition when the reactor is operating and makes the reactor vessel gas-tight in operation. When charging or discharging it is melted by electric heaters and becomes liquid and the plugs can then move and rotate. In addition to this freeze seal there are mechanical rubber seals.

The reactor has an axial and a radial shield. The radial shield is made of steel plate and concrete, the top shield consisting of steel plate and graphite. There are also thermal insulating layers. In spite of this protection, a certain quantity of heat is released, and this is absorbed by nitrogen cooling to ensure normal temperature for the concrete.

Such is the general design of the apparatus, but there are a number of fairly complex control mechanisms which are remote controlled. The remote control system provides for interlocks if a mistake is made by the operator. These interlocks ensure normal operation of all mechanisms and should provide protection against any kind of mistake.

Now something about the general mechanisms and how they differ from those used in England. Figure 1, Paper 3/1, shows the refuelling mechanism. It shows three positions: the first one is the non-operative position; the second is the approach to the assembly; and the third is the selection of assembly. I should like to stress that mechanisms have rigid kinematics; no chains or cables are used. We think that the kinematics should be rigid, otherwise the top part, i.e. the upper mechanism, could be operating and the actuating mechanism, the grab, could be in some unknown position and we would not know what was happening. Here in England it has been pointed out that cables and chains are used—parts which do not ensure

rigid kinematics in the mechanism. We never use cables in the refuelling mechanisms or in the safety and control mechanisms.

Figure 2, Paper 3/1, shows the elevator. There is a chain in it but this chain is in a guide and operates at both ends as a rigid link.

The BOR 60 (Fig. 14, Paper 3/1) is a small experimental reactor designed for a higher temperature. It has the same layout as the BN-350, i.e. sodium passed from the bottom to the top, and it also has a pressure chamber distribution header; the walls are sodium cooled. It has been designed for experimental purposes and for developing different fuel compositions, for testing different materials and fuel cans.

M. J. WHITMAN—*Paper 3/2*

Mr. Whitman summarized his paper and continued: Turning now to the Los Alamos Molten Plutonium programme, the initial interest in molten plutonium fuel arises from the potential for simplified fuel handling techniques. Current work on this fuel, however, has been concentrated on its use in a sealed capsule. Even in this conventional form, the liquid fuel retains several distinct advantages associated with the liquid state. That is, fission product formation cannot lead to mechanical straining of the fuel container. The liquid plutonium alloy would all be operated at such a temperature that phase changes of the fuel would not occur and thermal bonding and good heat transport are inherent.

The selection of a liquid metal alloy has further potential advantages. The hard neutron spectrum associated with a metallic fuel gives a high neutron multiplication ($n = 2.7$), thus suggesting the possibility of a very high gain breeder ($BR > 1.6$). Of equal or greater significance is the expansion coefficient associated with the metal. With the present container metals (Ta-5W is the reference container alloy), which have low expansion coefficients, the entire liquid volumetric expansion is effective in reducing core fuel density. Thus, a large, prompt negative temperature coefficient of reactivity is assured.

Figure D1 shows the present data on the time for various alloys of plutonium to diffuse through grain boundaries of a 0.020 in. (0.05 cm) tantalum wall, at various temperatures. It is clear that the ternary alloy, which is the present reference fuel in this programme, is far less subject to this diffusion than are unalloyed plutonium and the Pu-Fe eutectic. At temperatures of up to 750°C even this minute release of plutonium will not occur with the ternary fuel at times of interest for reactor applications. The dashed curve illustrates the improvement in the diffusion time for unalloyed plutonium fuel when carbon is diffused into the grain boundaries prior to introduction of the fuel. A similar change in slope by using carburization is expected for the Pu-Co-Ce alloy fuel case.

There has also been some experimental work on a niobium tungsten cladding that would be better from a neutronic standpoint than the tantalum clad, and the preliminary results are very promising.

In summary, a metallic alloy fuel, operated in the liquid state, has many features which make development of it attractive. Substantial effort has resulted in an apparently reliable fuel element. Safety features of this fuel

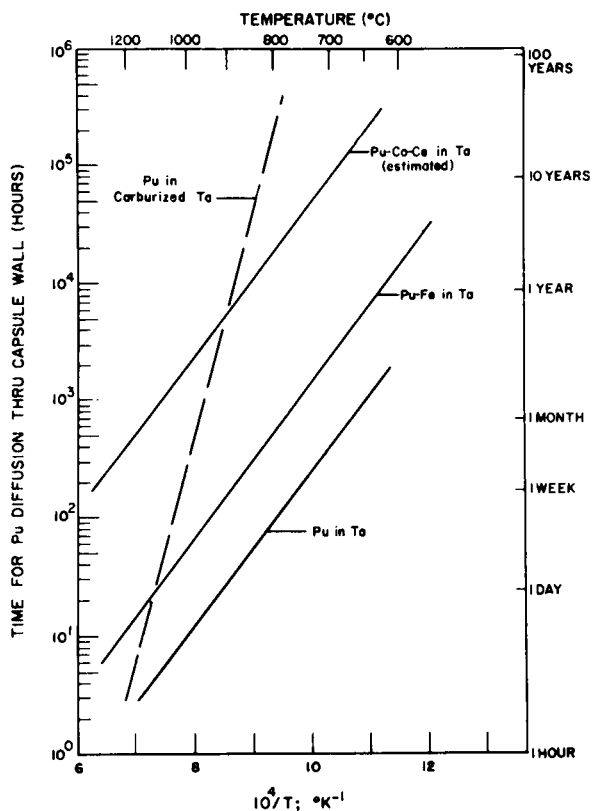


FIG. D1. Time for diffusion through capsule wall.

system are outstanding. The performance potential of this system, based on present technology, is attractive. Much work remains to be done in fully developing its potential.

A. G. FRAME—*Paper 3/3*

I am going to concentrate today on the configuration of the primary circuit of the PFR. The design of the prototype was preceded by an intensive study of commercial fast reactor power stations in the range 500–1000 MW(e). From these studies emerged a number of basic rules to which it was

felt the design of any sodium cooled fast reactor would have to conform at this point in time. These were:

- (a) that a secondary sodium circuit was necessary;
- (b) that boilers having single walls separating steam and sodium were essential for economic power reactors;

furthermore, the materials position was such that a once-through boiler could not be adopted; forced circulation would have to be used.

Turning to the core, we changed our minds on the direction of coolant flow from the Dounreay fast reactor and agreed that upward flow through the core was needed to guarantee heat removal at all times. The fuel subassembly should be designed for a 1000 MW(e) station with, of course, adjustment of enrichment to suit that core. Because of our interest in vented fuels it was felt that the system should be designed to cope with them. Our information was such that we had to design the gas blanket to contain the highly active gaseous fission products. We therefore concluded that it was advisable to have a low gas blanket pressure. The fuel pins in the subassembly, designed to contain fission product gases, should have a plenum to accommodate 100% of the fission product gas, and this plenum should be situated at the foot of the fuel subassembly to reduce the problems of bowing of subassemblies and to ease the problems of subassembly support.

Finally, on fuel handling, it was felt that the problems of cooling very highly rated plutonium subassemblies shortly after the reactor was shut down were such that we had to handle them under sodium until they could be removed from the reactor, not relying on forced circulation. We felt this was particularly important because undoubtedly defective fuel elements would have to be extracted from these reactors.

We felt that these ground rules were important and they have strongly influenced the design of PFR. Furthermore, when applying these ground rules to the design of the prototype we felt that it must be possible to uprate the solutions to a larger reactor—we did not want to adopt a unique solution for PFR. This is not to say, of course, that the solutions applied in PFR will not be improved on with time.

Conceptual designs of reactors conforming to these ground rules, particularly the ones on low gas blanket pressure and the fuel handling system, showed that an arrangement in which the reactor core heat exchangers and pumps were incorporated in separate vessels, interconnected with pipe work, was very space consuming. Latterly we were able to simplify this arrangement by adopting a support system like the Windscale AGR, with concentric inner and outer ducts—the hot coolant being in the inner duct. To give an idea of the space taken up by this arrangement on the 1000 MW(e) reactor, the minimum vault size that could be achieved was about 110 ft. For an arrangement in which the primary circuit was incorporated in a single tank the

equivalent figure was 60 ft. It was also found that in the single tank scheme there were many advantages because of the large volume of sodium inherent in the system; this helped to damp out some reactor transients. On the other hand, there were worthwhile operational advantages in the multi-tank system in that removal and maintenance of equipment was rather easier. However, pipe work decided us to opt for the single tank system. We did not have the necessary information to design high temperature pipe work up to 600°C for a 30 year life. We chose the very much simpler, from the sodium containment point of view, single tank arrangement.

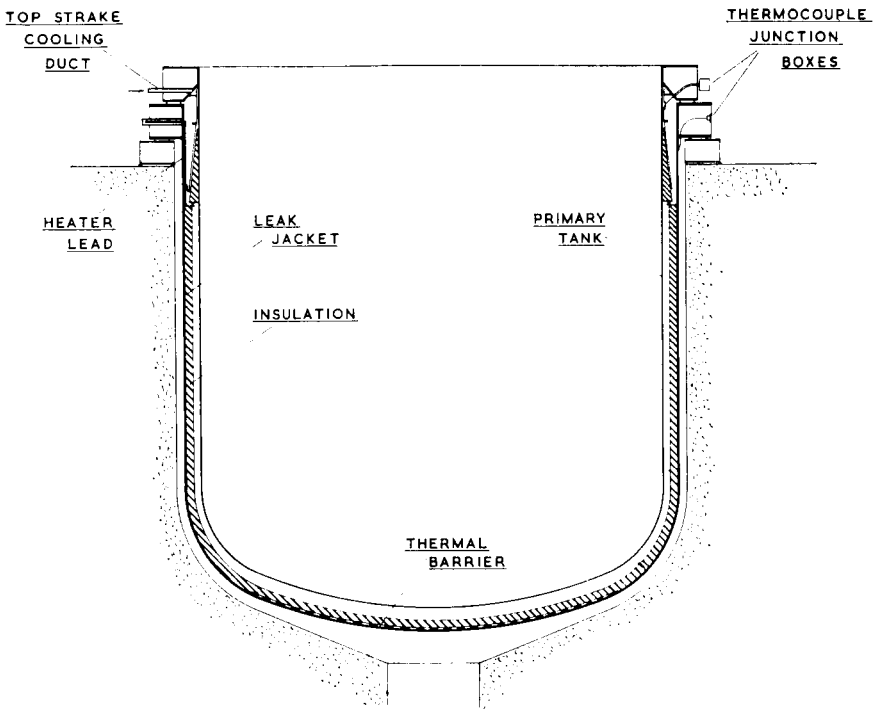


FIG. D2. Reactor tank and leak jacket.

What then has been our experience since we adopted this design (Fig. D2)? The design of the reactor tank and jacket shown here with the thermal insulation on the inside has presented few problems. It is a simple flanged vessel with no connexions or penetrations below the sodium level. The only real problem is in the top strake where there is a thermal strain induced by the expansion of the tank outwards owing to the high temperature sodium and the clamp flange at the top. It has been possible to overcome this with graded insulation and cooling on the top strake but suitable tanks can now be fabricated.

The design of the internal support structure has presented some quite difficult problems (Fig. D3). For instance, the roof, which carries a dead load of about 2000 tons, has some seventy-five penetrations for equipment and instrumentation. The detailed design of these penetrations with their thermal insulation and cooling system was not easy. Special internal insulation had to be developed for use in sodium circuits to keep the temperature of the reactor jacket down so that its load carrying capacity could be best used.

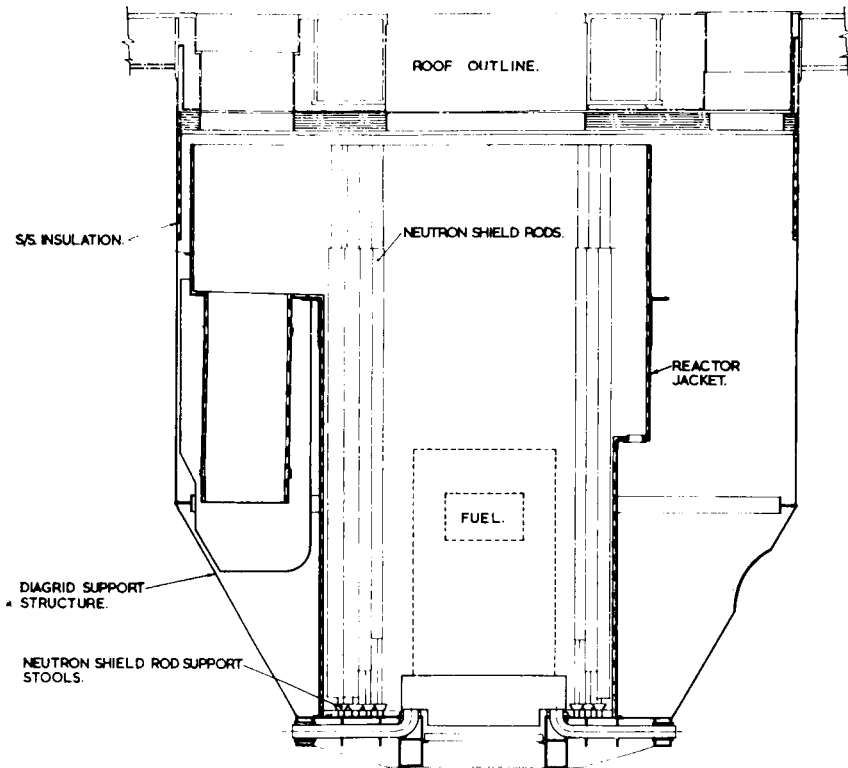


FIG. D3. Diagram of support structure.

Because of the integral nature of the design a very careful and detailed analysis of the problems of vibration and gas containment in the system has had to be made. The difficulties of designing the radial neutron shield are well known; there are problems of vibration, cooling, graphite expansion, and compatibility and finally of site construction. These are more difficult in the multi-tank scheme and have to be taken into account in the early stages of the design.

On our choice of primary circuit, we have now developed a design in great detail. We are absolutely convinced that this was the right choice and did

not lack any information. We can now design the system and feel that the choice made some months ago was correct.

In conclusion, I would like to review the status of the project programme. The design of the primary circuit, primary and secondary pumps, and turbo-alternator system has been completed and tender inquiries have been issued to industry. The design of the steam generator system and other major components is well in hand and contracts will be placed for all the equipment by the end of this year. The civil engineering contractor started on site on April 12, and is doing preliminary work. Excavation will start in July and finish at the end of 1966. Building work will start in 1967. The vessel contractor will move on to the site in February 1967. The reactor is scheduled to be on full power early in 1971.

L. VAUTREY—*Paper 3/4*

A few years ago, while the first fast reactor was being built in France, a study was made of fast reactors for the future. At the beginning of 1965 efforts were concentrated on the design study of the prototype reactor Phenix, the construction of which is expected to start in 1969. Phenix will be a sodium cooled reactor using mixed uranium and plutonium oxide for fuel. The power has not yet been defined, but studies are in progress on the reference values of 250 MW(e). This paper is devoted essentially to the design of the reactor and the cooling circuits. The various main elements of the circuits, such as pumps and steam exchangers, are detailed in Paper 5B/6.

C. A. PURSEL—*Paper 3/5*

[Mr. Pursel briefly described the four design studies of a 1000 MW(e) fast reactor published in January 1964, and continued]: The major conclusions from the studies were that with further studies and development it appeared to be feasible to design and operate an economically competitive 2500 MW(th) ceramic fuelled, sodium cooled fast reactor, and that there also appeared to be several practical ways to design a ceramic fuelled core with a negative power coefficient of reactivity.

It was recommended that future design studies should be directed towards developing a total plant concept in order to establish the research and development programmes required for these large reactors. Studies should examine the effect of core outlet temperature on plant cost; compare the fuel handling and transfer methods for reliability, short transfer time, and low cost; evaluate the single tank submerged primary system arrangement versus an external loop system; compare the conventional containment and a hot cell type structure; and define the areas that had the greatest potential for

reducing plant cost: assess the impact of private ownership of fuel on core and plant design; and further explore the effects of sodium voiding and the dynamics of coupled cores.

No study of the magnitude implied by these recommendations has been completed, but aspects of the problems have been studied and reported.

[Mr. Pursel then introduced the more recent studies described in his paper.]

J. O. TATTERSALL—*Paper 3/6*

For many years there has been close collaboration between British industry and the United Kingdom Atomic Energy Authority on the fast breeder project. As part of this continuing process the three industrial organizations, with which my co-authors and I are associated, were invited a little over 3 years ago to undertake a joint study of commercial sodium cooled fast breeder systems. This study—the major portion of which was completed last summer, and from which the contents of our paper are largely drawn—was timed to coincide with the formative stage of the prototype design, and had the dual objectives of producing firmer estimates of the economic incentive for the system and of ensuring, as far as is possible during the early stages of development, that the prototype and its associated development programme would provide adequate demonstration of all the important features that were likely to appear in the first commercial stations. One of the first features which had to be decided was unit size. Assuming 2 years' full power operation of the prototype, first inquiries for a commercial unit might be received by 1973, giving a possible commissioning date of 1978. It was estimated that by this time the interconnected grid in the U.K. would be large enough to accept units of 1000 MW(e). Today the estimate appears perhaps a little conservative, and 1200–1500 MW(e) might be nearer the mark. Some turbine manufacturers are currently thinking of single shaft machines of that size for the same period.

As far as costs were concerned, it was expected that in the U.K. competition would come from gas cooled or possibly water cooled thermal reactors, and extrapolation indicated that these types might be expected to give a generation cost of around $4\frac{1}{2}$ mills/kWh in a 2×1000 MW(e) station.

With regard to steam conditions, it seemed unlikely that the low fuel cost fast breeder would optimize at a higher temperature than the high fuel cost coal stations. For the latter there was no sign of any forthcoming breakthrough in either the cost or the high temperature properties of materials for the steam mains which would allow the steam temperature to be raised above the then favoured maximum of about 566°C. Accordingly, this figure was set as the target maximum for the sodium system, although it was suspected that the optimum might well be lower than that, perhaps 540°C. Recent information on 100,000 hour creep performance of the materials

suggests that this lower figure may now be the economic maximum, even for the coal stations.

In a double sodium circuit design of a fast breeder reactor, 540°C steam would imply a core outlet temperature of around 575–585°C, compared with the 600–610°C originally selected. As mentioned in the paper, early misgivings on materials behaviour and weld integrity contributed to the adoption of a Lamont type boiler for the reference design. Since that time confidence has been growing in stabilized 2½% Cr, 1% Mo steel and in the possible application of 9% and 12% chrome steels throughout the steam generators. As a result a new look is being taken at once-through sub- and supercritical designs.

Granted satisfactory materials, it has recently been suggested that savings of up to \$6/kW might be achieved on a 2×1000 MW station. It is perhaps worth recalling at this point that if operational proof of a different type of boiler is required provision for a fourth secondary loop has been made in the prototype layout.

With regard to fuel breeding, or rather doubling time, it is obvious that, if the plutonium fuelled fast breeder system is the most economic, then there will be an incentive to use it for as much of the future installation programme as possible. Viewed in this light, the performance of our reference design is perhaps barely sufficient, but, as indicated in the paper, substantial improvements can be made without significant effect on plant design or capital cost.

In the paper we quote a 6-year exponential doubling time for a modified fuel design with increased axial breeder. Current physics data would suggest that this figure is a little optimistic and perhaps it should be nearer 7½ years. Similar adjustments should be made to the figures quoted in ref. (4) of the paper.

When considering fuel handling a great deal of thought was given to the possibility of on-load charging because of the high cost of shutdown time associated with an outgoing station to make up station differential fuel costs from as much as 3.5 mill/kWh. However, this approach was eventually, if somewhat reluctantly, abandoned, and effort devoted to minimizing the down time associated with an off-load scheme. This, in turn, had some influence on the decision to increase the size of subassembly finally selected for the prototype, and on the selection, for the commercial station, of a fuel cycle geared to short duration refuelling shutdowns at weekends, when the demand is least, and hence the cost of replacement power somewhat lower.

Having selected the major features, the complete station design was worked out in detail and costed. This reference design is described in outline in section 4 of the paper. Since it was developed in parallel with that of the prototype important features are generally similar.

We have six pumps feeding sodium to a common plenum up through the core into a distribution tray and down through eighteen primary/secondary

heat exchangers. There is a rotating shield at the top. The in-vessel fuel handling machine transfers fuel from the core to the rotating decay store. It is perhaps worth noting that the roof structure is fairly massive, with large radial beams and a fairly large section box girder holding the main weight of the core. Another point to note on this arrangement is that we have allowed for variable gagging to the individual subassemblies in groups of two or three, and also to the breeder assemblies. This was perhaps one of the results of our earlier attempts to achieve 610°C mixed mean outlet without excessive cladding temperatures. The need for it is tied to both the outlet temperature and also the fuel cladding failure mechanism. We would hope, certainly with current thoughts on outlet temperatures, to simplify the core support structure and dispense with much of this individual gagging. However, we were anxious not to underestimate problems that might face us in the future.

A cross compound turbine was put in because we were more sure of the design and cost of these machines although we expect that, on the time scale for civil reactors, single shaft machines may well be used.

Turning finally to the costs, a detailed breakdown is given in section 3 of the paper. They relate to U.K. conditions and the generation costs have been worked out according to current U.K. ground rules.

As a result of our studies we are satisfied first that large commercial units, based on the prototype concept, are certainly feasible on the basis of current technology; and second, that there is every reason to expect that they will bring about a substantial reduction in the cost of nuclear power.

Discussion

H. M. CARRUTHERS (CEGB): I think one of the striking things about the way in which design concepts are emerging—at least in Western Europe—is the similarity of the solutions which are being favoured for prototypes by different countries. This would be very natural if it were obvious that only one combination of features would lead to competitive commercial fast reactors, but I am not sure that this is so. In these circumstances potential customers may well wonder whether there is a case for an even greater degree of international collaboration than exists at present. Such a change could be imagined as happening in one of two ways. It could be in a reduction in the number of prototypes which would otherwise be exploring virtually the same problems in the same way, and this would imply that the best choice of parameters and design features is indeed agreed to be well defined as of today, and it would involve international financing of the remaining project or projects on Dragon or Sefor lines. The other way would be for the various organizations to agree to minimize duplication of effort and deliberately to pursue different lines so that all potentially attractive

variants would be explored. This would imply that information between projects would have to be more liberally exchanged, including the traditionally more secret know-how aspects. Comments from the authors on this would be appreciated.

Now for two questions to which I should really like straightforward answers. Firstly, the special safety problems of fast reactors have appeared in several of the papers at this Conference, and the United States in particular have gone out of their way in designing fast reactor cores to minimize positive void coefficients. It would be interesting if we could have any guidance on the cost penalty which may be associated with these refinements.

Secondly, turning to the important question of plant reliability, it is encouraging to note that designers have arranged that the maximum number of components will be withdrawable for maintenance or replacement. It is clear though that, even with such a design philosophy, it is still vitally important to minimize component failure rates because of the high cost of replacement electricity during outages. In particular I would like to ask the authors of this morning's papers what design changes to steam raising units they would think most appropriate if currently proposed designs incur excessive loss of availability through leaks. Here one is thinking about, for example, a greater degree of subdivision in the boilers or an inert intermediate heat transfer medium. Double wall tubes might cost two or three times as much as single, but it is worth remembering that each 1% improvement in availability is going to be worth something like £1 per kW under U.K. conditions.

T. DOYLE (Euratom): I have two points. The first concerns the use of vented fuels in future fast reactors which I feel has not been adequately discussed at the Conference. If I may make some statements and ask for comments on them, I would say that there are perhaps three major aspects to the use of vented fuels. The first one is concerned with the clad performance, the second would be concerned with the safety of the reactor, and the third would be the overall economics. The clad performance was raised during the session on fuel technology, where it was asked: Should we develop new alloys for cladding or should we try to improve the ones we have? And the answer in general was that we would be far better off to try to improve the existing claddings. It seems to me that this is true but we should also follow a vented fuel development programme in case we do not manage to produce a cladding which is adequate for closed pin performance in future fast reactors.

On the safety aspect, the point of view has been defended by the United States in particular, and was mentioned again this morning in Paper 3/1, that vented fuels could contribute appreciably to the safety of fast reactors by avoiding gas blanketing, and so on, and this viewpoint has been contested especially during the last Argonne Conference by the British. I would say

that there must be a bit of stand-off between the two designs, and if anything the safety of the closed pin should not be put in doubt.

On the economics, some studies have been made and published, but the economics here, of course, embrace the fuel cycle cost and the capital cost of the plant, and here again there seems to be a stand-off between the two types with no appreciable gain, if any, for the vented design. Therefore, it seems to me that we would be well advised to develop vented fuels in case we cannot produce a clad which is adequate for future fast reactor service, but on the other two points we really cannot say that one is better than the other.

My second point is addressed to the Russian delegation. It was not clear from the presentation this morning, or from the paper itself if in the BOR reactor there is one or more closed loops for testing fuel. I put my question as follows: If there is such a closed loop in the reactor could we have a brief description of its characteristics and perhaps its design, and if there is no such closed loop in the reactor, to what extent can the BOR reactor establish the ultimate characteristics of the fuel in a manner which would not be given already by the BN-350 reactor?

J. F. W. BISHOP (UKAEA): My first questions are addressed to Mr. Whitman. In Paper 3/2 it is said that vented fuel is preferred for the fast ceramic reactor, but the general conclusion in earlier sessions was that sealed fuel was favoured for immediate development. Since fuel swelling is the major recognized failure mechanism, and we think by intergranular cracking of cladding in stainless steel clad designs, do the authors still support the selection of vented fuel? For what major reason, and what do they consider to be a realistic time scale for the introduction of vented fuel?

On gas cooled reactor fuel, I should like to ask what is the design concept of shaped pellets mentioned in the paper to minimize can/fuel interaction. And how does this concept bear on the swelling problem? Secondly, how does venting the fuel help in achieving better fuel element performance in the light of fuel swelling failures in a reactor which would have a high external pressure on the fuel?

Thirdly, on the molten plutonium reactor, have problems been encountered connected with fuel frothing and trapped fission products? Do the authors see any cladding material in sight which is compatible with the fuel and which has acceptable coolant corrosion for a modern steam cycle?

Then I have a question for Mr. Pursel. We have heard that 304 steel is compatible with a plutonium-uranium-zirconium alloy, as reported by Dragon workers, up to 800°C or so, and that high burnup is possible with strong cladding of dubious corrosion resistance. What promise is there of high burnup in a high temperature cycle?

Finally, if I may express our thoughts on vented fuel. We have included

vented fuel in our longer range development programme, as was indicated in one of our papers. Originally we included vented fuel for the usual reason, that we wished to ameliorate the major design problem of failure of sealed pins under internal pressure. We have not found failures on our programme under conditions relevant to PFR in a sealed design which we can attribute to internal pressure. However, we do intend—and indeed we shall be pushed by the engineers—to raise temperatures in the reactor to a point where ultimately we must expect failures caused by internal pressure, and it is for this reason that we are pushing on with vented fuel.

J. NIMS (APDA): I have two questions which I will address to Mr. Koshkin and Mr. Frame. The first question is concerned with the problem of sub-assembly vibration in free standing core subassembly designs. In the Russian paper the authors indicated that they had considered the effect of the hydraulic forces causing displacement of the subassemblies and estimated the static reactivity effects which could result from changes in flow. In addition to these flow dependent displacements there will also probably be some subassembly oscillation; if there is coupling between the subassemblies there will be reactivity oscillation and also power oscillation. Has this problem been considered by the designers?

The second question is concerned with the question of in-core instrumentation. There seem to be two schools of thought as to whether neutron detectors should be incorporated in the advanced core design. Those who feel it is not necessary rely on the use of outlet temperature measurements by thermocouples. The need for neutron detectors is probably more important in connexion with modular core design in large pancakes than it is for BN-350 and PFR. However, even for a reactor the size of Fermi we feel some neutron detectors would be extremely useful, particularly in connexion with the irradiation experiments which we have planned. Could we have the authors' comments on this question?

J. W. ASHLEY (U.K.): I really have no question to ask but some observations to make. I have heard various people say that they are concerned about the use of sodium. I think sodium is the only easy part of fast reactors and that the steam and water side has caused and will cause far more trouble than the sodium ever will. We have not yet seen, so far as I know, any steam generator fired by sodium, oil, water or helium, or any other material, which does not, in fact, cause considerable trouble through leakage, either into the steam or from the steam into the coolant. We should pay a little more attention, possibly, to that side of our engineering than to our core and its associated physics and design. Stress corrosion is a problem that has not yet been solved, and it may well be a very good cause for reducing our temperatures and coming back to some carbon steel steam generator of a

once-through type provided we face up to the stability problems that it will bring with it.

The second point concerns a comment made by Mr. Carruthers, when he suggested that various designs should be produced in various parts of the world and compared with each other. There are, in my experience, two things to compare: one is the systems that you are talking about; and the other is the organizations that are doing the work. As the difference in the organization is usually much bigger than the differences in the systems you are trying to compare, there is a great danger that you will end up by taking the system worked by the best team, and which may not be the best reactor.

I would like to deal in my third comment with the role of the design engineer. The design engineer, in my opinion, takes much too narrow a view. I believe he is responsible for seeing that the necessary research is done to support his work, that the design is worked out properly in conjunction with the people who are to manufacture the bits and pieces, that materials are not bought because they are cheaper but because they give the best overall quality, and that the quality assurance and inspection conditions are those intended by the designer and not worked out for some other organization. Then, most important, he should see that the construction is carried out in accordance with his plans and that the plant is commissioned. Many steam generators have never gone into operation because they have failed through stress corrosion during the time from which they left the manufacturers' works until they were due to operate. This almost undoubtedly occurs through the engineer not insisting upon adequate control right through to the construction stages.

L. M. WYATT (CEGB). I should like to raise points on Papers 3/1, 3/3 and 3/6 as they dealt more deeply with the aspects of fast reactors with which we, as operators, are most concerned. I shall start with Paper 3/3. This was concentrated on those parts of the unit with which operators have little trouble—the primary circuit generally—and tended to gloss over or miss the points where trouble does occur in operation. As far as the sodium cooled, fast reactor is concerned, the correct design, manufacture, and operation of the sodium heated steam generator is the most crucial part, and the authors of Papers 3/1 and 3/3 might well tell us a bit more about how they intend to overcome this difficulty.

As an incidental point, Mr. Frame said that materials prevented the use of a once-through steam generator. I would like more information on that point because I cannot see why they should.

The second point, referring solely to Mr. Frame, is that of fuel handling apparatus. The Russian contributors brought out very strongly the differences in the design of the fuel handling apparatus, which they obviously considered better than the British. It seemed to me that the difference was that they had

avoided using a pantograph and had divided up their fuel handling apparatus into two or really three parts. In our experience fuel handling apparatus gives a great deal of trouble, and pantographs do too, and I would like Mr. Frame's comments on whether the Russians really have found something better in design than he has.

The third point is that all our reactors have burst plug detectors, and we find this very useful. I do know that sodium does not attack the fuel in the way that carbon dioxide attacks the uranium in our magnox reactors, therefore burst plug detection, or some other means of detecting faults in the functioning of the fuel, is not quite so necessary, but if something comparable to a mechanic leaving a pair of pliers or a $\frac{1}{2}$ in. nut in the subassembly occurred, and one or more fuel elements melted, is there any way of getting an early warning before the whole core jams solid, or perhaps something worse happens? It seems to me that there is a possibility of something stopping coolant flow. Do we need any warning that this has happened before we get something really serious?

It seemed to me that the Russian fuel handling device lifted the fuel straight out of the sodium. Is the argon cooling good enough to take a fuel element immediately after reactor operation? In the British reactor you take the element out of the reactor core into sodium, and you have still got a good coolant and you can handle the fuel element fairly quickly after you have shut down the reactor, and therefore you save shutdown time. Is this a serious matter in the Russian reactor?

The last point is for Mr. Tattersall, who suggested that we fit a cross compound turbine or a very large turbine to the reactor. I think there is fairly general experience that reactors are more reliable than very large turbines. Might not there be some advantage in having, say, three turbines on his very large reactor?

D. SMIDT (Karlsruhe): What does Mr. Frame consider to be the maximum credible accident of the PFR? If it is connected with sodium ejection, what is the expected reactivity input rate and energy release?

My second question is to Mr. Whitman and stresses the point Dr. Bishop has already made. One of the alternative solutions of the gas cooled fast reactor is the collapsed canning concept. What is the expected change in coolant geometry by fuels running or by fuel creep; gas cooling is very sensitive, I think, to geometry. May it have any effect on heat transfer?

D. DONNA (Karlsruhe): I would like to ask Mr. Frame whether there could be an accident in which the flow of sodium in one subassembly were stopped completely. In this case the temperature of sodium would increase, and it could happen that the sodium would not boil at the saturation pressure, and you would have a certain amount of superheat. Unfortunately, when the

sodium is very pure, such as in fast reactors, and it is used at low pressure, such as in a fast reactor, the superheat could be very high—as high as 200°C. That means that when this energy stored in the superheated liquid is suddenly released—and one can calculate extremely short times, in the order of milliseconds—you would have an extremely quick ejection of sodium from the subassembly. If the subassembly is in the centre of the core surely you have a positive reactivity available in any reactor, and also in those reactors which have a coefficient for the total which is negative, but the trouble is not finished. It appears that this sudden release of energy produces increases of pressure which are such that could destroy the walls of the subassembly and thus propagate the failure.

T. P. HAIRE (CEGB): Paper 3/3 stated that kinetic studies showed that a two out of three failure of the primary sodium pumps is acceptable. I would like to know if a one out of two failure is acceptable in order that the reactor may be operated in the event of one primary pump being out of commission. Also, will the pumps be monitored for vibrational stresses during operation?

The second part of my question refers to the viability of metallic fuels. In view of the high coolant purity that is required for the current cladding materials associated with metallic fuels, is it considered possible to maintain the primary coolant in the PFR at the required levels of purity in the event of a break through in metallic fuels leading to the possibility of a metallic core?

M. J. RASTOIN (Fontenay-aux-Roses): My questions relate to Papers 3/3 and 3/4, both of which present single tank designs of reactors. First, about the shielding materials in the PFR design—clad graphite was chosen. Has one sufficient information on the behaviour of the cladding and of the graphite in these conditions for a discharge not to be necessary, and what would happen in the case of a break in the cladding?

My second question relates to the discharge of the diagrid. In the French design it is not considered necessary to discharge the grid. On the contrary, in the PFR design it would seem that the grid can be dismantled. I should like to know whether this operation, which seems to require at first sight a very important disturbance, has been considered absolutely necessary.

D. R. SMITH (TNPG): It is interesting to note the strong difference of emphasis in the approach to the safety question in the British and American designs presented today. The British approach can perhaps be summarized as follows: firstly, the probability of incidents leading to a major energy release in the core is made very remote by instrumentation and detailed

mechanical design; and secondly, the provision of a degree of containment appropriate to the low probability of this accident occurring. In contrast, the designs outlined in Paper 3/5 focus attention on a particular class of incident, namely sodium expulsion leading to core compaction and energy release, and the authors propose various core configurations—modular, annular and pancake—to minimize the consequences. These configurations will, as has already been mentioned, all involve some economic penalty. The possibility of other types of accidents leading to core energy releases is not discussed. I would like to hear the American authors' views on the British approach and, in particular, on the significance of the SPERT type incident referred to in section 4, Paper 3/3. This incident, I should explain, is the explosive generation of sodium vapour arising from a sudden contact with very hot fuel. Secondly, I would like their opinion on the reference in section 2 of Paper 3/6 to the low energy release now predicted from core compaction. Thirdly, how do the U.S. authors propose to deal with the containment for the very large energy releases referred to in their paper?

M. A. DE STORDEUR (Euratom): I would like to hear the British view on the compatibility of the tank type design with the idea of abandoning a secondary system one day in order to make the capital costs lower, and which would finally mean having water in this big tank. Also, what do the British think about the limit in size (in power) compatible with the tank type design?

D. OKRENT (ANL): The authors of Paper 3/6 do not give the reactivity coefficients and reactivity effects for their large reactor design. However, unlike many of the reactors described in Paper 3/5, no major modification of core geometry has been made to reduce positive sodium void effects. What criteria or philosophy will the authors use in assessing the acceptability of positive sodium void effects? What are the key questions, if any, which need to be answered in order to make large positive sodium void effects acceptable? Is any work at all planned on alternate core designs like the annular shape or the core made on many modules?

What is the current British philosophy for operating the prototype, or the eventual large scale reactor, in the event of fuel failure being detected? Is reactor operation expected to continue? What instrumentation do they feel is vital for this purpose? What are the key questions which remain to be answered? Have any criteria been established concerning acceptable behaviour of fuel in the event of fuel failure? Are any special measures in reactor plant design contemplated to allow for operation with failed fuel?

Finally, if the representatives from the U.S.S.R. find time to respond to Mr. Doyle's question concerning BOR 60, could they describe the reasons for the choice of open core tests in fuel testing rather than the use of many closed loops?

Authors' Replies

A. I. LEIPUNSKII—*Paper 3/1*: There was the very important point made that one of the weakest parts of the whole nuclear power plant was the steam generator, and what would we do if the present design were not to work. For the steam generator, the only solution is a very high quality of work at all stages—good tubes, good workmanship, and very good methods of fault detection at every stage of production. Although especially true for steam generators, this also applies to all equipment for fast nuclear reactors cooled by sodium.

There was then a question about vented fuels. We are all interested in them, and know that, from the point of view of radioactivity, it is quite possible to work with vented fuels and nothing very much happens, but we still prefer to start our work with sealed fuels; because of some improvements, it is easier for maintenance. However, I still think that we shall develop vented fuel even if it will only give some higher reliability, because there are always various small factors which can distort neutron flux and energy generation, so all the security possible is needed for fuel, the most important part of the reactor.

Then there were some questions about BOR 60 experimental plant—why didn't we have loops, and if not why don't we arrange them? We want to commission this reactor as soon as possible, so we have tried to make it as simple as possible. A loop is very complicated to maintain so we decided not to have any but to have some subassemblies in which to put the instruments, so there will be instruments positioned in the core to know what happens there.

In reply to the question about vibration and stability in our BN 350, we have tried seven sub-assemblies in a water loop and have taken measurements along the whole length of one of them. We know what to expect.

A very important question referred to in-core instrumentation. Physicists would like to have every pin instrumented; the design engineer would like to have no instrumentation at all. We have to draw a line somewhere—I think nearer to the design engineer.

A few words about the charge/discharge machine—I think there is some misunderstanding. Around the outside, the breeding region of the reactor, we have holes for the subassemblies, and we wait about three months to cool them in sodium, but outside the reactor.

Lastly, I should like to say something about the maximum accident. I think we have two philosophies here, one mainly in various European countries, and the other in America. In America, the design is from the point of view of the sodium void coefficient, and therefore the core is divided, or made like a very thin pancake. Although in Europe this is taken into account, I think the British view is rather more economic and perhaps the right one to adopt.

M. J. WHITMAN (USAEC)—*Paper 3/2*: I would first like to answer the question about the vented fuel concept—its timing and the statement that it is the preferred reference for the fast ceramic reactor. It is just that it is preferred by the General Electric Company. As a matter of timing, we have a strong R&D programme under way on fuels, and there is time for this decision to be made. For the present it has been chosen as the preferred reference fuel. The reasons for wanting to go to the vented fuel concept are generally well recognized. The gradual release of fission gases will prevent sudden bursts that could cause blanketing of adjacent fuel elements, and perhaps a very serious accident. The evidence that has been obtained shows that not only can the short lived fission gases be held up within this venting mechanism, but that the long term fission products also are held up, so that you do not have a really serious problem with contamination, which is encouraging.

There is also a possibility that the actual size of the venting device would be smaller than that required for the pressure containment. There is also the possibility of thinning the cladding. One point brought out was that this does not necessarily take care of the swelling and rupture of the cladding. However, some of the slides gave evidence that you could successfully operate with molten core fuel, and there is a very good possibility that the fuel temperatures can be high enough for the plasticity of the fuel to take care of this swelling problem.

In addition to that, we also have an intensive R&D programme under way on fuel cladding, so that we may have to handle that in the same way that our British and European friends are doing.

On the questions dealing with the gas cooled concept, I would like to call upon my co-author, Dr. Fortescue, whom I believe also has some comments on the vented concept.

P. FORTESCUE (General Atomics)—*Paper 3/2 cont.*: There may have been a slight misunderstanding in the paper in that it may have been taken that the remarks on voided cans related equally to sodium and to gas systems. This is certainly not the case. Quite different considerations affect the gas cooled fast reactor. Whereas it is reasonable to consider venting fission products to a sodium primary circuit on the grounds that the activity is already there anyway and it is a double isolation, to release fission products to the primary circuit of a gas cooled reactor would be a much more serious step to take. In the gas cooled system the idea of vented cans is regarded as very much a last ditch approach, and, if we were driven to it, it would not be voiding to the coolant but voiding to an external vent, which, fortunately, is rather more easy to engineer in the case of the gas system than in the sodium system by virtue of the characteristically smaller number and larger diameter of the fuel elements. The straight answer is that the voided, vented fuel is not recommended at the moment for the gas cooled system.

Dr. Bishop raised some specific points about the specially shaped fuel pellets. The paper merely referred to dishing of the ends and the provision of a centre hole in the pellets designed to match the dimensional changes of the pellet surface with temperature more closely with those of the surrounding can. It is an attempt, in other words, to control thermal ratcheting. It is one of the few practical points about the system that has been fairly fully demonstrated so far because it is comparatively easy to do a large number of thermal cycles early in the experimental programme. In fact, some of the capsules irradiated in Oak Ridge have undergone a very large number of thermal cycles with no evidence of thermal ratcheting at all. We attribute this largely to the detailed design of the pellet shape.

Dr. Smidt raised the question of fuel swelling and how it was dealt with. Here, of course, one must distinguish between the so called fuel supported type of can and the flex can design mentioned in the paper. The latter is a sodium filled system with a pressure equalization mechanism in the shape of a somewhat deformed void section at the end of the can. In the latter case you use the highest possible fuel density and you allow about 20% for volume expansion in the space between the can and the pellets. There is no problem there from the growth of the can. It does not have to support pressure. In the case of the fuel supported type of fuel element, the word is a little misleading. It is fuel supporting in the sense that at the hot spots the external gas pressure can, by a process of creep, force the can on to the fuel, but in the cooler regions of the fuel element it is not fuel supported. The point of the distinction is this, that it is in the hottest regions that you have the least amount of swelling to allow for in the oxides because there the fission products tend to be less retained; in the cold regions is where you have to worry badly about swelling, and so our approach is that in the so called collapsed clad types of fuel element we provide a 20% volume in the pellets by way of using an 80% density. This is actually distributed in the form of about 85% density fuel with a 5% hole in the middle, and in the cooler regions of the core, where you might get a 20% volume increase, the can is strong enough to retain it. In the hotter regions of the core we can, in fact, easily take 2 or 3% increase in can diameter without greatly causing malfunction of the cooling, but we would be much more worried by the ductility of the can and by shutting of the coolant.

A. G. FRAME—*Paper 3/3*: Dealing first with the steam generator question, which was asked by several people. We have not underestimated the problems of the steam generator. We have put a significant design and development effort into this steam generator and we deliberately chose not to change our design to a once-through arrangement because our design and development was far advanced and we did not want to fall into the pitfall of having had insufficient development on a new steam generator.

There are two types of systematic faults which can be caused by vibration—fretting and maldistribution of coolant. We feel we can solve these by good design. The other type of fault is in fabrication—tube to tube welding, and high quality tubes can only be achieved by a significant improvement in the standards currently available in industry all over the world. We have spent a lot of money in this country, and we feel that in the PFR steam generator we will demonstrate that this can be done. If I were to hazard a guess as to what one would do if it could not be done, I would turn people's attention to the EBR-II steam generator, where there is a duplex tube and double tube plate. I do not think we can afford to pay the price, but it looks a reasonable solution.

Mr. Nims asked a question on subassembly vibration. We have considered this, and in our paper (and Paper 5B/5) you will see that we have studied vibration, and have solved the vibration problem in PFR. There was no reactivity effect through vibration of subassemblies. For in-core instrumentation, we are providing a delayed neutron monitoring location system in PFR. Thermocouples help, but thermocouples over every subassembly are of little use in determining small defects in pins. They are useful for the operators but do not significantly improve safety.

With regard to fuel handling, I can only say that the Russian fuel handling system looks excellent, but we do not propose to change ours. We have put considerable effort into the reliability of the fuel handling equipment and have given very careful attention to the detailed design and are doing development work on those areas with which we are concerned.

As far as the BCD system is concerned, I would only point out that we do test the subassemblies for blockage by flow before we put them into the reactor. We have filters on the inlet of every subassembly and we feel our only problem is blockage that can come from within.

Dr. Smidt asked about the maximum credible accident. This is indicated in the paper. We regard the propagation of subassembly faults as the most pressing problem facing the fast reactor designer today as far as core design is concerned, and we have assumed that we cannot detect certain types of faults, although it is very unlikely that these will happen, and have designed our containment system on this basis. We are running these subassemblies in the Dounreay fast reactor and they are behaving perfectly normally.

The question from Mr. Donne on superheat is rather related to the SPERT type accident referred to in the paper, where we have assessed the energy release that is available in a very short time scale, and this is the accident that worries us most on fast reactors at the present time. We think superheat is very unlikely when you have got a blocked subassembly and particulate matter and gas present to break up the possibility of superheat.

In reply to Mr. Haire, we can operate with one out of two pumps. We have

to keep the temperature rise over the core the same, and therefore this means a reduction of power. We do not propose to use hot traps in PFR.

There was a question from M. Rastoin on shielding materials. We feel that we have sufficient information to design the graphite steel structures forming the radial neutron shield, and are obtaining information on what happens if we have a defective element, and are designing the system to restrain any expansion that might be caused. The shield rods are, of course, removable.

Mr. de Stordeur asked about tank design—we would not put heat exchangers with water inside the large tank.

Mr. Okrent asked about current philosophy in operating with defective fuel elements. My opinion at this stage is that when we begin PFR we will tend to take subassemblies out when we get a fault, although subsequent experience between now and the time that PFR operates might cause us to change this view. Our biggest concern is the rate of propagation of the fault. This is the big uncertainty in subassembly behaviour at the present time.

L. VAUTREY—*Paper 3/4*: As from the end of 1967 three prototype steam generators will be tested, and, as far as techniques are concerned, we have chosen a design of steam generator that includes modules, that is to say, each element is of a reduced power, and among the three prototypes or designs or models of steam generators that will be tested, the one which represents the greatest power per module is 60 MW. By module, I include the evaporator, the re-heater and the superheater. In this type, which was studied at CEA, the various tubes in the reactor come out of the envelope and there was no welding inside the apparatus. In the various models that we have studied and among the criteria that we have adopted, we have paid special attention to problems of safety and risks of leakage, or consequence of such leakage.

C. A. PURSEL—*Paper 3/5*: I would only comment on the safety discussion that, as it says in the paper, the cores that were shown were all grossly distorted, probably overly so, to accommodate the void coefficients. It is usually assumed that these represent overly spoiled geometries, and that as more experience is gained, more knowledge accumulated, and more discussions have taken place, the pancakes will no longer be quite so pancaked, and the modular cores will no longer have so many modules, and so they might ultimately all look the same, and the number of modules might be reduced to one.

As far as design considerations are concerned for the core compaction accident, I might refer to the SEFOR designs since they have just gone through it.

Now to deal with timing vented versus non-vented fuels—to design engineers in general, it appears not to make any difference whether you design for one or the other. The design considerations are about the same—

that is outside the core—since if you have non-vented fuel there will be a continual pressure to go to higher and higher burnups and to higher and higher temperatures, to the point where you begin to have fuel failures, so you begin to cope with problems other than the sudden burst anyway.

I would also like to say something about metallic fuel elements which have been mentioned several times. Typical ones are 19% plutonium, 14% zirconium; a lead one is now in ZPR-V, where the fast to thermal flux ratios are 6 to 5. It is now at 7.8 atm.% burnup, and accumulating burnup at the rate of about 0.7% per month. The fuel has a sodium bond. The gap will allow 30% radial expansion. The cladding is vanadium titanium alloy. Since it is in a capsule, the environment of the cladding is stagnant NaK. There are other capsules which have not gone quite as far in ZPR-V and EBR-II. The EBR-II capsules are full length, but they too are encapsulated. It is a safety requirement for EBR-II that screening tests must be encapsulated. The maximum burnup in EBR-II is 2.2%. Some of these capsules are encapsulated in 304, some 316, and some in Hastelloy, and the metallurgists feel that these may well turn out to be compatible with both the sodium and the metal.

Written Contributions

J. O. TATTERSALL—*Paper 3/6*: In reply to Mr. Wyatt, the projected use of a single large turbine was aimed purely at economy and I agree that if the current operating experience with turbines were to be a continuing feature, then there might well be an economic advantage in a multiplicity of smaller units. However, I believe that many of the present troubles have arisen because the turbine manufacturers have scarcely had time to digest the very rapid advances in design that have taken place in the last 15–20 years. To have to install, by the late 1970's, machines twice the size of those being installed today, but having the same steam conditions, does not seem to me to involve the same pace of development and this should be reflected in increased reliability.

With reference to M. de Stordeur's question, some attention was given to investigating larger sizes during the course of study. In broad terms it was concluded that the engineering of units up to three times the output—the largest considered—would be quite feasible. The vault roof structure could become difficult, but, if so, there is no real need to retain a round tank, although the adoption of another shape might aggravate the tank support problem.

So far as single sodium circuit systems are concerned, it is not at all certain, if and when these are developed, that the large low pressure single tank system would still be used, since this concept was evolved specifically with a two circuit system in mind. At present it does not seem very likely that the

possibility of a violent sodium–water reaction could be entirely eliminated with any known steam generator design and it would be imperative to isolate the core from damaging shock waves and significant quantities of reaction products. A great deal more information has to be gathered about the detection, nature, and consequences of sodium–water reactions. When this has been done and the new design criteria placed in perspective with the old, it may well be (if single sodium circuits are shown to be acceptable) that the overall design concept will become very different.

Dr. Okrent notes that we do not appear to have made modifications to the core geometry to reduce the reactivity effects of sodium voidage. This is quite true, although during the formative stages, and indeed throughout the study period, a broad range of safety studies and allied topics undertaken by UKAEA were closely followed, and these included assessment of the various reactivity coefficients for large cores of the type we are proposing. At the same time, we were aware that in the United States considerable attention was being devoted to core designs which minimized positive sodium void coefficient.

Our philosophy has been—and still remains—that large sodium voids must be prevented, not principally because of the void reactivity coefficient, but more because of the potentially much greater hazard that could arise from melt down and subsequent core compaction. The design of the system has to be such that the sudden development of a large void or the simultaneous development of a significant number of small voids both have a negligible probability and that the safety system must be capable of detecting the onset of any local fault and shutting the reactor down before any serious propagation has occurred. In this context, the sodium coefficient has a possible modifying effect on the size/duration aspect of faults that can be tolerated before shutdown, but it is only one of many criteria involved and changes in design aimed at its reduction would not necessarily result in an overall improvement. Within the time scale of the study it was felt that any attempt to achieve a merit comparison of alternative geometries would not be justified having regard to the uncertainties in the data available.

With regard to the particular projected design of the civil reactor presented in our paper, the general intention is to retain as closely as possible the design of fuel subassemblies proved in the prototype. This means that the core of the 1000 MW(e) civil reactor will have four times the number of subassemblies and an L/D ratio smaller by a factor of 2. Comparative estimates of void coefficients prepared by UKAEA indicate that the reactivity effect of voiding a subassembly in the civil reactor would be some 40% of the corresponding effect in the prototype. The general nature and consequences of faults are, of course, continually under review, but at present there are no plans for modifying the core geometry so as to alter the sodium coefficient.

As far as the operation with failed fuel is concerned, an assessment of the extent to which this can be tolerated in the civil reactors will have to await results of the UKAEA development programme. It is not expected that the detection of small quantities of fuel in the coolant will be required as an indication of some impending major fault demanding immediate shutdown, but it is as yet too early to be sure of this. The economic pressure to raise the tolerance level will be determined by the frequency of failures and this, in turn, could be significantly affected by the successful development of vented fuel designs.

N. G. WORLEY (Babcock & Wilcox (Operations) Ltd.): There were a number of general questions at the Conference on boiler design and reliability with sodium heating. As designers and manufacturers of steam raising plant, we feel that some comments on this subject are timely. We were associated with the UKAEA in the general development of the designs which are being adopted for the prototype fast reactor plant and have been used as a basis for the commercial fast reactor study described in Paper 3/6. Throughout these studies it has been the guiding principle that, wherever possible, features which could lead to troubles should be eliminated. This required a careful survey of the manufacturing processes, erection procedures, and the effect these can have on the eventual design. The arrangement, therefore, uses simple U-tubes with no tube welds in contact with sodium, the tubes have a limited amount of manipulation with large radius bends. The selection of a forced circulation design ensures that, as long as the circulation ratio is adequate, no part of the tubing in contact with sodium will be subjected to thermal cycling or alternate wetting and drying which will unavoidably be the case of once-through plant. Again, the design has been selected to enable replacement of faulty tube bundles to be carried out reasonably quickly and the selection of the number of units in the commercial design was in part governed by the balance between initial capital cost and the size of unit for replacement. It is, of course, intended that supervision of manufacture at all stages from the tube making to the completed unit will include all the techniques that are available to ensure that no faulty material is used in the construction of the plant.

It is our general opinion that reliable plant can be built by taking precautions of this type and it is probable that some of the design criteria that we have used could be relaxed as development work demonstrates more conclusively the limits of operation of the materials being used, both in sodium and with steam-water mixtures.

While it is possible to take these general precautions, there still remain the practical problems of ensuring that all stages of a complicated fabrication, particularly while producing plant on a large scale for a large power producing programme, are carried out at the standards that are necessary. It is

also likely that, while prototype plant operation may be satisfactory, standards of operation associated with a large power programme may not be so high, leading to serious plant failures.

Experience with the magnox reactor boiler plant has been that, while the Calder and Chapelcross boilers have had only minor tube leakage problems, some of the later Electricity Board stations have had serious leakages of water into the gas circuits.

The consequences of tube failures with the steam generators can be serious, both in the cost of repairs and outage time, and we do not believe that it is possible to assess the magnitude of this problem. We feel, therefore, that it is essential for parallel work to be carried out on concepts other than the use of sodium as a secondary circuit fluid. Mr. Woodcock, in an independent contribution to the discussion, has suggested that a gas-solid mixture has attractive possibilities as an alternative.

Author's Replies

W. H. HANNUM—*Paper 3/2*: Dr. J. W. Bishop asked if there was any problem with frothing of the fuel or with trapped fission products. The answer is: no—with the liquid plutonium fuels of current interest, there has been no evidence of formation of froth. The volatile fission products retained in the fuel rapidly reach saturation under operating conditions at about 4–6 vol.% of the fuel. This quantity is stable and offers no operational problem. The process of fission product release is as follows: the volatile fission products nucleate on available sites on the capsule walls and grow to a release size of 1 mm in diameter. The bubbles then break from the walls and rise through the liquid breaking free on the surface. The equilibrium of 4–6% by volume corresponds to saturation of available nucleation sites.

Dr. Bishop also asked if there were any clad materials which were compatible with the molten plutonium fuels which also have acceptable sodium corrosion performance properties. In this case the answer is: yes—the tantalum alloys, as described in the paper, are fully adequate for containment of the liquid plutonium alloys of interest at reactor temperatures or above. Tantalum alloys are also fully acceptable in sodium, providing reasonable care is exercised in sodium purity. This is illustrated by the almost three years of operation of LAMPRE, in which many samples of tantalum (the fuel elements) were exposed to flowing sodium under reactor conditions. There was no evidence of corrosion of the tantalum by the sodium. A great number of sodium loops containing these fuel elements have also been run, with no detectable corrosion of the tantalum or tantalum alloy capsules by the sodium. Niobium may also be acceptable, but the testing to date is inadequate to establish a reasonable level of confidence.